

UD-7429

**Integrated M. C. A. (Sem. - II)
Examination, June-2025**

IMCA - 204T

Graph Theory and Discrete Structures

Time Allowed : Three Hours

Max. Marks : 70

Mini. Marks : 28

This question paper contains two sections as under:

Section-A**Max. Marks-20**

This section will have 10 questions, having 02 questions from each unit with no internal choice. Answer in about 20 words. All questions will have equal marks. (10×02=20)

Section-B**Max. Marks-50**

This section will have 10 questions, having 02 questions from each unit, one question to be attempted from each unit. Words limit approximately 400-500 words. Questions may have sub divisions. All questions will have equal marks. (10×05=50)

SECTION – A

- 1
- (a) Define bipartite graph and cut edges.
 - (b) Define hamiltonian graph and graph coloring.
 - (c) Define planar graphs.
 - (d) Define directed tree and spanning tree.
 - (e) State Pigeon Hole principle.
 - (f) State principle of Inclusion-Exclusion.
 - (g) Define Generating function.
 - (h) Define Linear recurrence relations with example.
 - (i) State principle of duality for lattices.
 - (j) Define distributed lattices.

SECTION - B

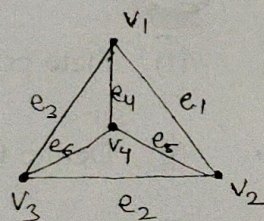
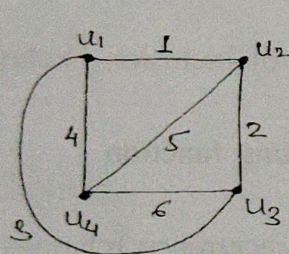
UNIT - I

- 2 (a) Show that the number of odd vertices in a graph is always even.

(b) The sum of the degrees of all vertices in a graph G is equal to twice the number of edges in G .

OR

- 3 (a) Show that the graphs given below are isomorphic :



- (b) A simple graph with n -vertices and k components can have at most

$$\frac{(n-k)(n-k+1)}{2} \text{ edges. Prove it.}$$

UNIT - II

- 4 Prove that a given connected graph G is an Euler graph iff all vertices of G are of even degree.

OR

- 5 Let G be an acyclic graph with n vertices and k connected components i.e. $w(G) = k$, then G has $(n - k)$ edges.

UNIT - III

- 6 (a) Suppose 14 students in a class appear at a test. Prove that there exist at least two among them where seat numbers differ by a multiple of 13. (2)
- (b) Out of 1000 students, 750 offer Hindi and 400 offer English. How many can offer Hindi only and how many can offer English only?

OR

- 7 For any finite sets A, B, C , prove that

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C) \quad \text{here } n(A) \text{ is cardinality of } A.$$

UNIT - IV

- 8 (a) Find generating function for the Fibonacci sequence $\{a_r\}$ defined by

$$a_r = a_{r-1} + a_{r-2}, \quad r \geq 2, \quad a_0 = 0, \quad a_1 = 1.$$

- (b) Find $c = a * b$ if $a_r = 3^r, r \geq 0$ and

$$b_r = 2^r, r \geq 0.$$

OR

- 9 Solve the difference equation

$$a_r - 4a_{r-1} + 4a_{r-2} = (r+1)2^r$$

UNIT - V

- 10 Let N^+ be the set of positive integers and the relation ' $\leq$ ' in N^+ be i.e. x/y be " x divides y ". Show that N^+ is a lattice where the join ($\vee$) and meet ($\wedge$) are resp. defined as

$$a \vee b = LCM(a, b) \quad \text{and} \quad a \wedge b = HCF(a, b)$$

OR

- 11 (a) In a distributive lattice, if an element has a complement then this complement is unique.
(b) The dual of a complemented lattice is also a complemented lattice.